

Abstract

Intrusion Detection Systems (IDS) play a crucial role in protecting computer networks from cyber-attacks. Traditional IDS systems rely on signature-based detection methods, which are ineffective against new and unknown attacks. To overcome these limitations, machine learning techniques can be used to automatically identify malicious activities in network traffic. This project, Smart Shield: ML-Based Intrusion Detection System, uses machine learning algorithms to detect network intrusions and classify traffic as normal or malicious. The system uses publicly available datasets such as UNSW-NB15 and NetFlow datasets to train the detection model. The system employs algorithms like Random Forest and Decision Tree to analyze network features and detect suspicious patterns. The developed system processes network traffic data, performs feature extraction and preprocessing, and uses trained machine learning models to detect attacks. The system provides accurate detection results and helps security analysts identify potential threats in real time. The proposed IDS improves detection accuracy and reduces false alarms compared to traditional systems. This solution can be used in enterprise networks, cloud environments, and cybersecurity monitoring systems to strengthen network security.

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Chapter 1: Introduction

1.1 Background

With the rapid growth of the internet and digital technologies, cyber-attacks have become increasingly common. Organizations rely heavily on computer networks to store and transfer sensitive information. However, these networks are vulnerable to attacks such as malware, denial-of-service attacks, and unauthorized access.

To protect networks from such threats, Intrusion Detection Systems (IDS) are used. IDS monitor network traffic and detect suspicious activities. Traditional IDS use rule-based detection methods, which cannot effectively detect new or unknown attacks.

Machine learning techniques have emerged as powerful tools in cybersecurity. By analyzing patterns in network traffic data, machine learning models can identify anomalies and detect malicious behavior. This project focuses on building a machine learning-based IDS that improves the accuracy and efficiency of attack detection.

1.2 Intrusion Detection Systems (IDS)

An Intrusion Detection System is a security tool that continuously monitors network or system activities to detect unauthorized access or malicious behaviour. IDS systems generate alerts whenever suspicious activity is identified, allowing administrators to take necessary actions.

Types of IDS

IDS can be broadly classified into the following types:

1. Network-Based Intrusion Detection System (NIDS)

NIDS monitors network traffic across multiple devices and analyses data packets flowing through the network. It is usually deployed at strategic points such as routers or gateways.

2. Host-Based Intrusion Detection System (HIDS)

HIDS operates on individual devices and monitors system-level activities such as file access, system calls, and application logs.

3. Hybrid IDS

Hybrid IDS combines both network-based and host-based approaches to provide a more comprehensive security solution.

1.3 Need for Advanced Intrusion Detection

Traditional intrusion detection systems are primarily based on signature-based detection techniques, where known attack patterns are stored in a database. While these systems are effective in detecting previously known attacks, they have several limitations:

- Inability to detect unknown or zero-day attacks
- High dependency on frequent updates of signatures
- Increased false positive rates
- Lack of adaptability to evolving threats

As cyber-attacks become more advanced, traditional IDS systems struggle to keep up with new and sophisticated attack techniques. This creates a need for more intelligent and adaptive security solutions.

1.4 Role of Machine Learning in Intrusion Detection

Machine Learning (ML) is a subset of artificial intelligence that enables systems to learn patterns from data and make decisions without being explicitly programmed. ML techniques have gained significant importance in cybersecurity due to their ability to analyze large volumes of data and identify hidden patterns.

In intrusion detection, machine learning can be used to:

- Analyse network traffic data
- Identify patterns of normal and abnormal behavior
- Detect unknown and evolving attack types
- Improve detection accuracy
- Reduce false alarms

By training models on labeled datasets, machine learning algorithms can automatically classify network traffic as normal or malicious.

1.5 Types of Machine Learning Techniques Used

Several machine learning techniques are used in intrusion detection systems, including:

Supervised Learning

In supervised learning, the model is trained using labeled data where the output is known. It is commonly used for classification tasks in IDS.

Unsupervised Learning

Unsupervised learning identifies patterns in unlabeled data and is useful for anomaly detection.

Ensemble Learning

Ensemble methods combine multiple models to improve prediction accuracy and reduce overfitting.

In this project, supervised learning techniques such as Random Forest and Decision Tree algorithms are used to build the intrusion detection model.

1.6 Datasets Used in Intrusion Detection

Machine learning models require high-quality datasets for training and evaluation. In this project, datasets such as UNSW-NB15 and NetFlow are used.

- UNSW-NB15 Dataset contains realistic network traffic with various types of

attacks.

- NetFlow Dataset provides summarized network traffic information useful for detecting anomalies.

These datasets include multiple features that describe network behavior, enabling the model to learn patterns effectively.

1.7 Problem Statement

Traditional Intrusion Detection Systems (IDS) are primarily based on signature-based and rule-based detection techniques, where known attack patterns are stored and matched against incoming network traffic. Although these systems are effective in detecting previously known attacks, they face several major limitations. They are unable to detect unknown or zero-day attacks, which do not have predefined signatures. Additionally, they require continuous manual updates of attack signatures, making them inefficient and time-consuming to maintain.

Another significant challenge with traditional systems is the high rate of false positives and false negatives. False positives occur when normal network traffic is incorrectly classified as malicious, while false negatives occur when actual attacks go undetected. These issues reduce the reliability and effectiveness of intrusion detection systems in real-world environments.

Furthermore, with the increasing complexity and volume of network traffic, traditional IDS systems struggle to process large datasets efficiently. Modern cyber attacks are becoming more sophisticated, dynamic, and adaptive, making it difficult for static systems to keep up with evolving threats. This creates a need for more intelligent and automated solutions that can learn from data and adapt to new attack patterns.

To address these challenges, there is a need to develop an advanced intrusion detection system that leverages machine learning techniques. Machine learning algorithms can analyze large volumes of network traffic data, identify hidden patterns, and automatically classify activities as normal or malicious. Such a system can improve detection accuracy, reduce false alarms, and adapt to new types of attacks without requiring manual updates.

Therefore, the problem is to design and implement a machine learning-based intrusion detection system that can efficiently analyze network traffic, learn from historical data, and accurately detect both known and unknown cyber attacks, thereby enhancing overall network security.

1.8 Objectives

The main objective of this project is to design and develop an efficient and intelligent Intrusion Detection System (IDS) using Machine Learning techniques to identify and prevent unauthorized access and malicious activities in a network environment.

The specific objectives of the project are as follows:

- The system aims to identify various types of cyber-attacks such as Denial of Service (DoS), Probe attacks, User to Root (U2R), and Remote to Local (R2L) attacks by analysing network traffic patterns.
- The project utilizes Machine Learning models such as Decision Tree, Random Forest, or other classifiers to distinguish between normal and malicious network

behaviour effectively.

- A key objective is to minimize false positives (normal traffic classified as attack) and false negatives (attacks not detected), thereby improving the reliability of the system.
- The system is designed to automatically monitor network traffic and detect threats without requiring constant human intervention.
- The project focuses on handling large datasets and extracting meaningful patterns that can help in identifying suspicious activities.
- By detecting intrusions in real-time or near real-time, the system contributes to strengthening the overall security of the network infrastructure.
- The proposed system should be capable of adapting to new types of attacks by updating or retraining the machine learning models.

1.9 Scope of the Project

The scope of this project defines the boundaries and extent to which the system is developed and implemented. This project focuses on the application of Machine Learning techniques for network intrusion detection in a controlled environment.

The scope includes the following aspects:

- The system analyses network traffic data to identify patterns associated with normal and malicious activities. It focuses on features such as protocol type, duration, service, and number of connections.
- The project utilizes standard datasets such as KDD Cup 99 or NSL-KDD for training and testing the machine learning models. These datasets contain labelled instances of both normal and attack traffic.
- The system is limited to implementing selected machine learning algorithms for classification purposes. Deep learning or advanced hybrid models may not be included in the current scope.
- The project primarily supports offline analysis of network data and may include basic real-time detection capabilities depending on implementation.
- The project is implemented as a software prototype and does not involve deployment in a real-world enterprise network environment.

Chapter 2: Literature Survey

Several research works have been conducted in the field of intrusion detection systems using machine learning. Many traditional IDS systems use signature-based detection techniques, which depend on known attack patterns. These systems fail to detect new or unknown attacks. Researchers have proposed machine learning based IDS models that analyses network traffic patterns and automatically detect anomalies. Algorithms such as Decision Tree, Random Forest, Support Vector Machine, and Neural Networks have been widely used for intrusion detection. Studies have shown that Random Forest algorithms provide higher accuracy and better performance in detecting attacks compared to other models. Random Forest uses multiple decision trees to improve prediction accuracy and

reduce overfitting. Datasets such as KDD Cup 99, NSL-KDD, and UNSW-NB15 are commonly used for evaluating IDS systems. These datasets contain various types of network attacks including DoS, Probe, and malware traffic. However, many existing IDS solutions still face challenges such as high false positives and limited scalability. Therefore, this project proposes a machine learning-based IDS that improves detection performance and supports multiple datasets.

2.1 Previous Research

Intrusion Detection Systems (IDS) have been widely studied in the field of cybersecurity to detect malicious activities in computer networks. Many researchers have explored the use of machine learning techniques to improve intrusion detection accuracy and reduce false positives.

One of the earliest datasets used for IDS research is the KDD Cup 1999 dataset, which helped researchers develop several machine learning based detection models. However, due to redundancy and outdated attack patterns, newer datasets such as NSL-KDD and UNSW-NB15 were introduced.

Several research studies have applied machine learning algorithms such as Decision Tree, Random Forest, Support Vector Machine (SVM), Naive Bayes, and Neural Networks for detecting network intrusions. Among these algorithms, Random Forest has shown high accuracy and better generalization capability for detecting complex attack patterns.

Researchers have also explored ensemble learning techniques, where multiple machine learning models are combined to improve detection performance. These techniques help reduce overfitting and increase reliability in intrusion detection.

Recent studies have focused on improving IDS by using deep learning methods and real-time traffic monitoring systems. However, implementing these systems requires high computational resources and complex infrastructure.

2.2 Existing Systems

Before the introduction of Machine Learning-based Intrusion Detection Systems, traditional methods were used to secure networks and detect malicious activities. These systems mainly relied on predefined rules, signatures, and manual monitoring techniques to identify potential threats. The existing systems can be broadly classified into Signature-Based Detection Systems and Anomaly-Based Detection Systems.

1. Signature-Based Intrusion Detection System

Signature-based IDS detects attacks by comparing network traffic against a database of known attack patterns or signatures. If a match is found, the system flags it as a potential intrusion.

A well-known example is Snort, which uses predefined rules to identify malicious packets.

Working Principle:

- Maintains a database of known attack signatures
- Monitors incoming network traffic

- Compares traffic patterns with stored signatures
- Generates alerts when a match is found

Advantages:

- High accuracy for known attacks
- Easy to implement and understand
- Low false positive rate

Limitations:

- Cannot detect new or unknown (zero-day) attacks
- Requires frequent updates of signature database
- Ineffective against evolving attack pattern

2. Anomaly-Based Intrusion Detection System

Anomaly-based IDS works by establishing a baseline of normal network behaviour and detecting deviations from this baseline as potential threats. These systems often use statistical methods or basic machine learning techniques to identify unusual patterns.

Working Principle:

- Learns normal network behaviour
- Monitors real-time traffic
- Detects deviations from normal patterns
- Flags anomalies as potential intrusions

Advantages:

- Can detect unknown and novel attacks
- Useful for identifying insider threats
- Adaptive to changing network behaviour

Limitations:

- High false positive rate
- Requires training phase
- Difficult to define “normal” behavior accurately.

2.3 Comparison with Proposed Approach

The proposed system, **Smart Shield: ML-Based Intrusion Detection System**, improves upon the limitations of existing systems by using machine learning algorithms to detect network intrusions automatically.

Unlike traditional IDS that rely on predefined signatures, the proposed system learns

patterns from network traffic data and can detect both known and unknown attacks.

The system uses UNSW-NB15 and NetFlow datasets for training machine learning models such as Random Forest and Decision Tree. These algorithms analyse multiple network features and identify suspicious patterns with higher accuracy.

Compared to existing systems, the proposed system offers the following advantages:

- Ability to detect unknown attacks
- Higher detection accuracy
- Reduced false positive rates
- Automated learning from datasets
- Improved scalability for large network environments

By combining machine learning techniques with network traffic analysis, the proposed system provides a more intelligent and efficient solution for intrusion detection.

Chapter 3: System Analysis

3.1 Existing System

Traditional intrusion detection systems use rule-based or signature-based detection methods. These systems monitor network traffic and compare it with known attack signatures.

Examples include:

- Snort
- Suricata
- Signature based IDS tools

3.2 Problems in Existing System

The existing intrusion detection systems, which mainly rely on signature-based and anomaly-based techniques, face several challenges in effectively securing modern network environments. With the rapid growth of cyber threats and increasing network complexity, these traditional systems are no longer sufficient. The major problems associated with the existing systems are discussed below:

3.2.1 Inability to Detect Zero-Day Attacks

One of the major limitations of traditional intrusion detection systems is their inability to detect new or unknown attacks. Signature-based systems depend on predefined attack patterns stored in a database. If a new type of attack occurs that does not match any existing signature, the system fails to detect it.

This makes the system vulnerable to zero-day attacks, which are newly discovered vulnerabilities exploited by attackers before any preventive measures are developed.

3.2.2 High False Positive Rate

Anomaly-based intrusion detection systems often generate a large number of false positives. Normal user activities may sometimes deviate slightly from the predefined behavior, causing the system to incorrectly classify them as malicious.

This leads to:

- Unnecessary alerts
- Increased workload for administrators
- Difficulty in identifying actual threats

As a result, the efficiency of the system decreases significantly.

3.2.3 High False Negative Rate

In some cases, the system may fail to detect actual attacks and classify them as normal behavior. This is known as a false negative.

False negatives are more dangerous because:

- Real attacks go undetected

- Network security is compromised
- Sensitive data may be exposed

3.2.4 Lack of Learning and Adaptability

Traditional systems do not have the ability to learn from new data or past experiences. They operate based on fixed rules and require manual updates to incorporate new attack signatures.

This lack of adaptability makes them ineffective against:

- Evolving attack techniques
- Dynamic network environments
- Sophisticated cyber threats

3.2.5 Scalability Issues

With the increase in network size and traffic volume, existing systems face difficulties in processing large amounts of data efficiently. As the number of users and devices grows, the performance of these systems degrades.

This results in:

- Slower detection
- Increased processing time
- Reduced system efficiency

3.3 Proposed System

To overcome the limitations of traditional intrusion detection systems, this project proposes an advanced Machine Learning-based Intrusion Detection System (IDS). The proposed system is designed to intelligently analyse network traffic, detect malicious activities, and classify them as normal or attack using various machine learning algorithms.

Unlike existing systems, the proposed model does not rely solely on predefined rules or signatures. Instead, it learns from historical data and identifies patterns associated with both normal and malicious behaviour. This makes the system more adaptive, accurate, and efficient in detecting both known and unknown cyber threats.

3.3.1 Objectives of the Proposed System

The proposed system aims to:

- Improve the accuracy of intrusion detection
- Detect both known and unknown attacks
- Reduce false positive and false negative rates
- Automate the detection process
- Handle large-scale network data efficiently
- Provide a scalable and adaptable security solution

3.3.2 Working of the Proposed System

The proposed system follows a structured workflow to detect intrusions effectively. The major steps involved are:

3.3.2.1 Data Collection

Network traffic data is collected from standard datasets such as KDD Cup 99 or NSL-KDD. These datasets contain labeled records of both normal and attack traffic.

3.3.2.2 Data Preprocessing

The collected data is cleaned and prepared for analysis. This includes:

- Removing irrelevant or duplicate data
- Handling missing values
- Converting categorical data into numerical form
- Normalizing or scaling features

3.3.2.3 Feature Selection

Important features are selected from the dataset to improve model performance and reduce complexity. This step helps in removing unnecessary attributes that do not contribute to intrusion detection.

3.3.2.4 Model Training

The processed data is used to train multiple machine learning algorithms, including:

- K-Nearest Neighbors (KNN)
- Random Forest
- Support Vector Machine (SVM)
- XGBoost

Each algorithm learns patterns from the training data and builds a model for classification.

3.3.2.5 Model Testing

The trained models are tested using unseen data to evaluate their performance. Metrics such as accuracy, precision, recall, and F1-score are calculated.

3.3.2.6 Classification

When new network traffic data is provided, the system classifies it as:

- Normal traffic
- Malicious (attack) traffic

3.3.2.7 Result Analysis

The performance of each algorithm is compared to determine the most effective model for intrusion detection.

3.4 Key Features of the Proposed System

3.4.1 Intelligent Detection Mechanism

The system uses machine learning algorithms to automatically identify patterns and detect intrusions without relying on predefined rules.

3.4.2 Multi-Algorithm Approach

By implementing multiple algorithms (KNN, Random Forest, SVM, XGBoost), the system ensures better performance and allows comparison between models.

3.4.3 High Accuracy and Efficiency

The use of advanced algorithms improves detection accuracy and reduces false alarms.

3.4.4 Adaptability

The system can be retrained with new data, allowing it to adapt to evolving cyber threats.

3.4.5 Scalability

The proposed system can handle large datasets and is suitable for modern network environments with high traffic volume.

3.4.6 Automation

The entire detection process is automated, reducing the need for manual intervention.

3.5 Advantages of Proposed System

- The proposed Machine Learning-based Intrusion Detection System offers several significant advantages over traditional intrusion detection methods, making it more suitable for modern network security requirements. One of the primary advantages is its ability to detect both known and unknown attacks. Unlike signature-based systems that rely on predefined attack patterns, the proposed system learns from historical data and identifies patterns associated with malicious behavior. This enables it to effectively detect zero-day attacks and previously unseen threats, thereby enhancing overall network security.
- Another important advantage is the improvement in detection accuracy. By utilizing advanced machine learning algorithms such as K-Nearest Neighbors (KNN), Random Forest, Support Vector Machine (SVM), and XGBoost, the system can classify network traffic more precisely. These algorithms analyze multiple features and patterns in the data, reducing the chances of misclassification. As a result, the system achieves lower false positive and false negative rates, ensuring that genuine threats are detected while minimizing unnecessary alerts.
- The proposed system also provides automation in the intrusion detection process. Traditional systems require continuous manual updates and monitoring, whereas the machine learning-based system automatically learns from data and updates its model accordingly. This reduces the dependency on human intervention and minimizes maintenance efforts. Additionally, the system is capable of handling large volumes of network data efficiently, making it highly scalable for real-world applications where network traffic is continuously increasing. Overall, the proposed system enhances network security by providing intelligent, accurate, and scalable

intrusion detection. Its ability to learn, adapt, and automate makes it a reliable solution for protecting modern network infrastructures from a wide range of cyber threats.

Chapter 4: System Design

4.1 System Architecture

The system architecture defines the overall structure and workflow of the proposed Machine Learning-based Intrusion Detection System (IDS). It illustrates how data flows through different components of the system, from data collection to final prediction of network intrusions. The architecture is designed to ensure efficient processing of network traffic data, accurate detection of malicious activities, and scalability for handling large datasets. The system consists of multiple modules such as data collection, preprocessing, feature selection, model training, and classification.

4.2 Data Flow

4.2.1 Data Collection Module

This is the first stage of the system where network traffic data is collected. The dataset used in this project includes standard datasets such as NetFlow or UNSW-NB15, which contain labelled records of normal and malicious activities.

The collected data includes various features such as:

- Protocol type
- Duration of connection
- Source and destination bytes
- Number of failed login attempts

This module acts as the input layer of the system.

4.2.2 Data Preprocessing Module

The raw data collected from the dataset may contain noise, missing values, or irrelevant information. Therefore, preprocessing is necessary to prepare the data for machine learning models.

The preprocessing steps include:

- Data cleaning
- Handling missing values
- Encoding categorical variables
- Feature scaling and normalization

This step improves the quality of data and enhances model performance.

4.2.3 Feature Selection Module

In this stage, the most relevant features are selected from the dataset. Removing unnecessary or redundant features helps in:

- Reducing computational complexity
- Improving model accuracy
- Avoiding overfitting

Feature selection ensures that only important attributes are used for training the model.

4.2.4 Model Training Module

In this module, the preprocessed data is used to train multiple machine learning algorithms. The algorithms used in this project include:

- K-Nearest Neighbors (KNN)
- Random Forest
- Support Vector Machine (SVM)
- XGBoost

Each model learns patterns from the training data and builds a classification model.

4.2.5 Model Testing and Evaluation Module

After training, the models are tested using unseen data to evaluate their performance. Various evaluation metrics are used, such as:

- Accuracy
- Precision
- Recall
- F1-score

This module helps in selecting the best-performing algorithm for intrusion detection.

4.2.6 Classification Module

In this stage, the trained model is used to classify new incoming network traffic. The system predicts whether the traffic is:

- Normal
- Malicious (attack)

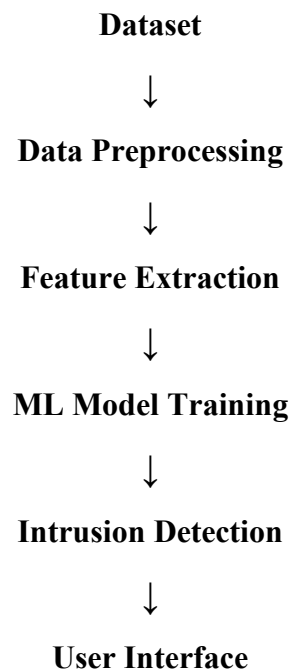
This is the core functionality of the intrusion detection system.

4.2.7 Result and Alert Generation Module

The final output of the system is displayed in this module. It shows:

- Detection results
- Type of attack (if detected)
- Performance metrics

4.3 Flow Chart



4.4 Block Diagram

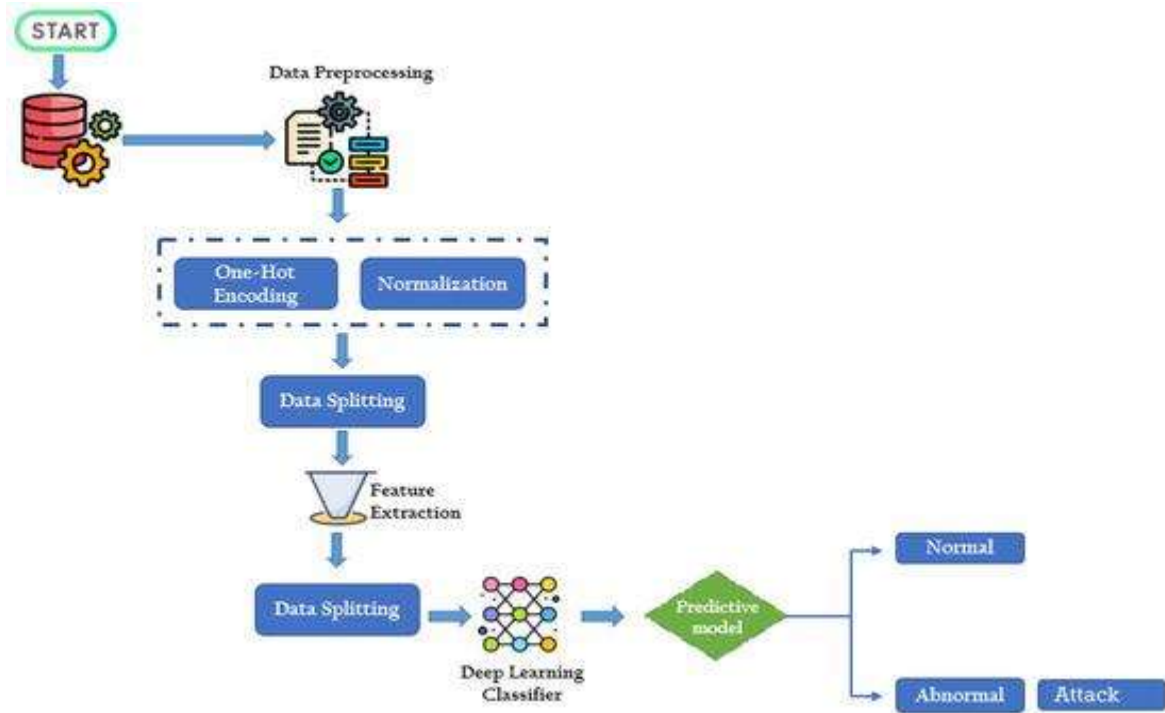


Fig 1: Block Diagram of Intrusion Detection System

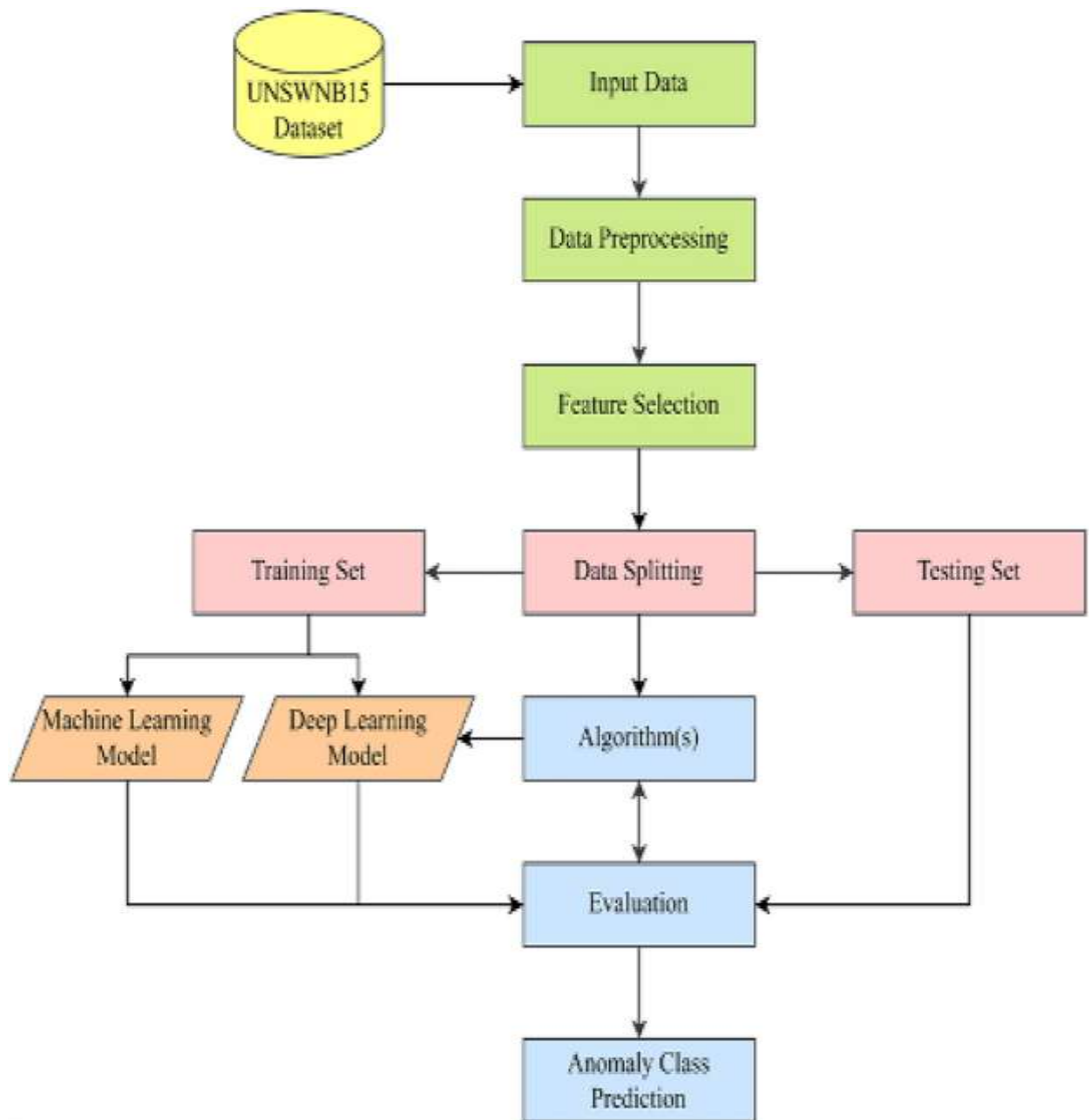


Fig 2: Signature Based Intrusion Detection using Machine Learning

4.5 Use-Case Diagram

se Case Diagram

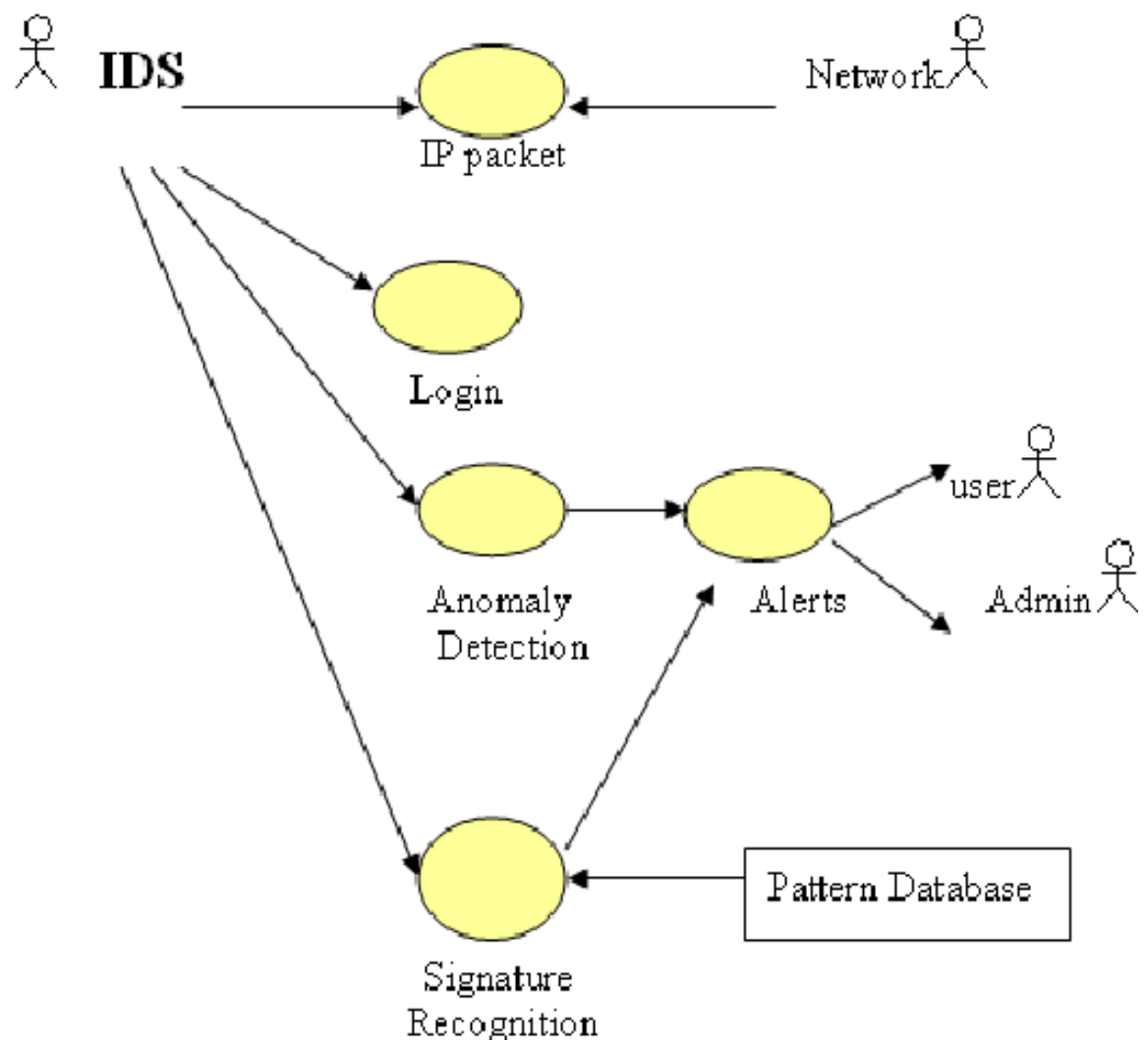


Fig 3: Use-Case Diagram of Intrusion Detection System

4.6 Data Flow Diagram

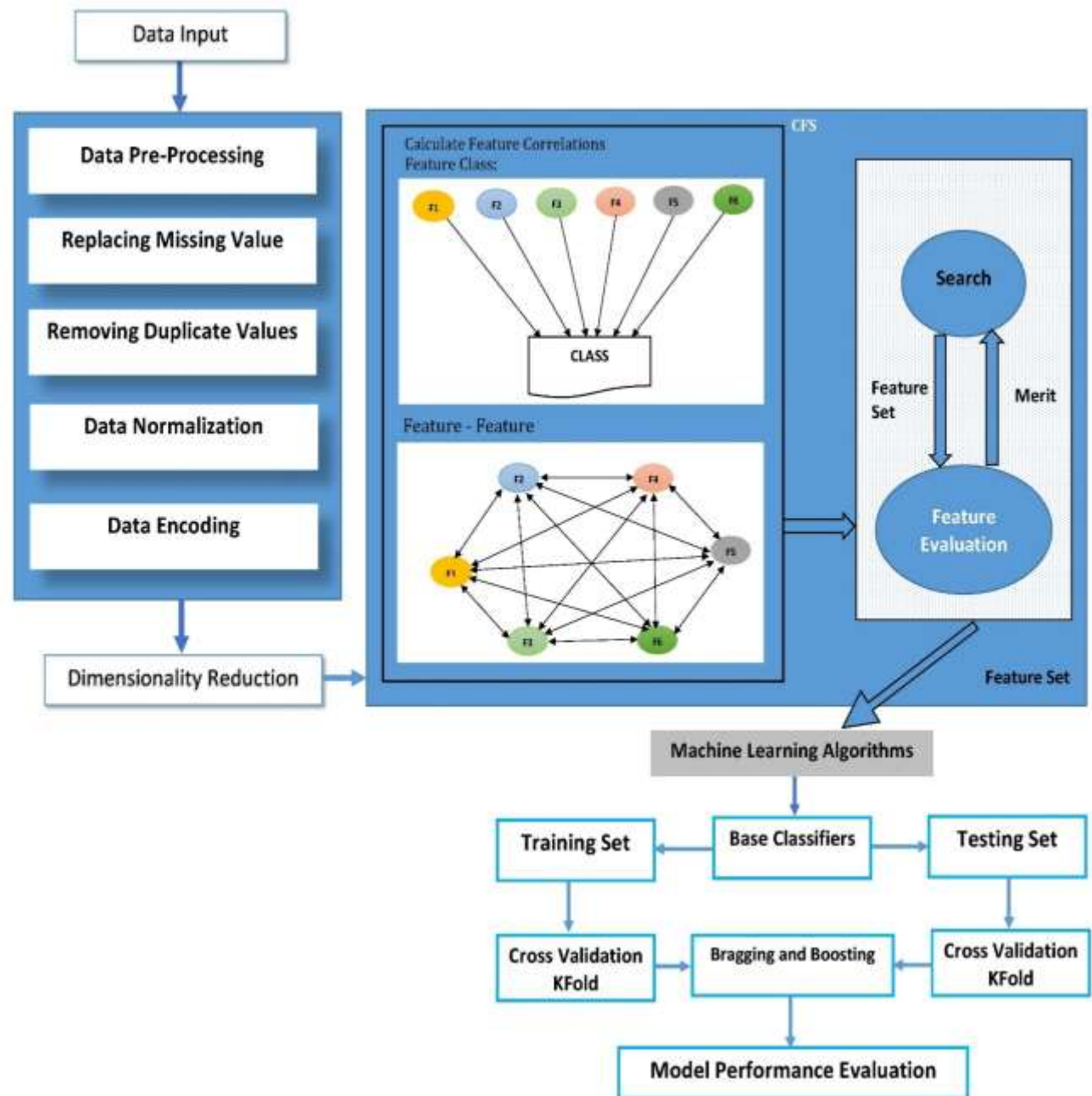


Fig 4: Data-Flow Diagram of Intrusion Detection System

Chapter 5: Implementation

5.1 Hardware Requirements

- Processor: Intel i5 or above

- RAM: 8 GB minimum
- Storage: 256 GB

5.2 Software Requirements

- Python
- ReactJS
- NodeJS
- Scikit-Learn
- Pandas
- NumPy
- Streamlit / Web UI
- VS Code

5.3 Algorithms Used

In this project, multiple Machine Learning algorithms are implemented to improve the accuracy and reliability of the Intrusion Detection System (IDS). The use of different algorithms allows comparison of performance and helps in selecting the most effective model for detecting network intrusions. The algorithms used in this system include K-Nearest Neighbors (KNN), Random Forest, Support Vector Machine (SVM), and XGBoost.

5.3.1 K-Nearest Neighbors (KNN)

K-Nearest Neighbors (KNN) is a simple and effective supervised machine learning algorithm used for classification tasks. It works based on the concept of similarity between data points.

In this project, KNN is used to classify network traffic as either normal or malicious based on the similarity of its features with the training data. When a new data point is received, the algorithm identifies the ‘K’ closest data points (neighbors) from the training dataset and assigns the class that is most common among them.

KNN is particularly useful in intrusion detection because similar attack patterns tend to have similar feature values. However, the performance of KNN depends on the choice of ‘K’ value and can be computationally expensive when dealing with large datasets.

Advantages:

- Simple and easy to implement
- Effective for small to medium-sized datasets
- No training phase required

Limitations:

- High computation time during prediction
- Sensitive to irrelevant features and noise

5.3.2 Random Forest Algorithm

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve classification accuracy and reduce overfitting.

In this project, Random Forest is used to classify network traffic by constructing multiple decision trees during training. Each tree gives a prediction, and the final output is determined by majority voting. This approach enhances the robustness and accuracy of intrusion detection.

Random Forest is highly effective in handling large datasets and selecting important features, making it suitable for network intrusion detection tasks.

Advantages:

- High accuracy and robustness
- Reduces overfitting compared to a single decision tree
- Handles large datasets efficiently

Limitations:

- More complex than individual models
- Requires more computational resources

5.3.3 Support Vector Machine (SVM)

Support Vector Machine (SVM) is a powerful supervised learning algorithm used for classification and regression tasks. It works by finding the optimal hyperplane that separates data points of different classes.

In this project, SVM is used to distinguish between normal and attack traffic by mapping input data into a higher-dimensional space and finding the best boundary that separates the classes.

SVM is particularly useful in high-dimensional datasets, which is common in network intrusion detection systems where many features are involved.

Advantages:

- Effective in high-dimensional spaces
- Works well with clear margin of separation
- Memory efficient

Limitations:

- Not suitable for very large datasets

- Choosing the right kernel can be difficult.

5.3.4 XGBoost Algorithm

XGBoost (Extreme Gradient Boosting) is an advanced ensemble learning algorithm based on gradient boosting. It is designed for high performance and efficiency.

In this project, XGBoost is used to improve the detection of complex attack patterns by combining multiple weak learners (decision trees) into a strong predictive model. It builds trees sequentially, where each new tree corrects the errors of the previous ones.

XGBoost is widely used in machine learning competitions due to its speed and accuracy, making it highly suitable for intrusion detection systems.

Advantages:

- High accuracy and performance
- Handles missing data effectively
- Efficient and scalable

Limitations:

- More complex to understand and tune
- Requires careful parameter tuning

5.4 Dataset Description

5.4.1 UNSW-NB15 Dataset

The UNSW-NB15 dataset is a modern network intrusion detection dataset developed by the University of New South Wales (UNSW), Australia. It was created to overcome the limitations of older datasets such as KDD-99 and NSL-KDD.

The dataset contains both normal network traffic and various cyber-attack activities. It was generated using the IXIA PerfectStorm tool in a controlled laboratory environment to simulate realistic network traffic.

The dataset includes multiple types of attacks such as:

- DoS (Denial of Service)
- Fuzzers
- Backdoors
- Exploits
- Reconnaissance
- Shellcode
- Generic attacks
- Worms

Each record in the dataset represents a network connection with several features describing the behavior of the traffic.

Key Characteristics

- Contains 2.5 million network records
- Includes 49 network traffic features
- Provides labeled data for both normal and attack traffic
- Suitable for machine learning based intrusion detection research

In this project, the dataset is used to train and test machine learning models for detecting malicious network activity.

5.4.2 NetFlow Dataset

The NetFlow dataset contains summarized information about network traffic flows. NetFlow is a network protocol developed by Cisco that collects metadata about IP traffic passing through routers and switches.

Instead of capturing full packets, NetFlow records information such as:

- Source IP address
- Destination IP address
- Source port
- Destination port
- Protocol type
- Number of bytes transferred
- Duration of the connection

NetFlow data helps analyze communication patterns within a network, making it useful for detecting suspicious activities such as abnormal traffic behavior, scanning attacks, or denial-of-service attacks.

Key Characteristics

- Represents network flow records instead of full packets
- Captures traffic statistics and communication patterns
- Efficient for monitoring large-scale network environments
- Useful for anomaly detection and attack detection

In this project, NetFlow data is used to train machine learning models that identify malicious network traffic based on flow characteristics.

5.4.3 NSL-KDD Dataset

In this project, the NSL-KDD dataset is used as the primary dataset for training and testing the Machine Learning-based Intrusion Detection System (IDS). The NSL-KDD dataset is an enhanced version of the KDD Cup 1999 dataset and is widely used in cybersecurity research for evaluating intrusion detection techniques.

The dataset was developed to address the limitations of the original KDD dataset, such as redundancy and imbalance, which negatively affected model performance. By removing duplicate records and providing a more balanced distribution of data, the NSL-KDD dataset ensures better training efficiency and more reliable evaluation of machine learning models.

Structure of the Dataset

The NSL-KDD dataset consists of a collection of network connection records. Each record represents a network session and is described using 41 input features along with one output label indicating whether the connection is normal or an attack.

Input Features

The 41 features describe various properties of network traffic, including:

- Duration of connection
- Protocol type (TCP, UDP, ICMP)
- Service type (HTTP, FTP, etc.)
- Flag status
- Number of bytes transferred
- Login and authentication information
- Traffic statistics

These features provide sufficient information to identify patterns associated with normal and malicious behavior.

5.5 Feature Categories

The features in the dataset are broadly classified into three categories:

1. Basic Features

These are derived from packet headers and include:

- Duration
- Protocol type
- Service
- Source and destination bytes

2. Content Features

These features are extracted from the data content and include:

- Number of failed login attempts
- Root access attempts
- File operations

3. Traffic Features

These features capture traffic patterns over a period of time:

- Number of connections to the same host
- Number of connections to the same service
- Error rates and connection statistics

Output Classes

Each record in the dataset is labeled as either:

- Normal (legitimate network activity)
- Attack (malicious activity)

The attack class is further divided into four major categories:

- Denial of Service (DoS): Attacks that overwhelm system resources
- Probe: Surveillance attacks to gather information
- Remote to Local (R2L): Unauthorized access from a remote system
- User to Root (U2R): Unauthorized access to root privileges

5.6 Features Used in the Project

Features are the attributes or variables that describe network traffic behavior. Machine learning models analyze these features to determine whether the traffic is normal or malicious.

Some commonly used features in the datasets include:

Network Connection Features

- duration – Duration of the network connection
- protocol_type – Type of protocol used (TCP, UDP, ICMP)
- service – Network service on the destination
- flag – Status flag of the connection

Traffic Statistics Features

- src_bytes – Number of bytes sent from source to destination
- dst_bytes – Number of bytes sent from destination to source
- count – Number of connections to the same host
- srv_count – Number of connections to the same service

Flow Features (NetFlow)

- source_ip
- destination_ip
- source_port

- destination_port
- protocol
- packet_count
- byte_count

These features help the machine learning algorithms analyze patterns in network traffic and identify abnormal behavior that indicates a cyber-attack.

Feature Name	Description
duration	Length of network connection
Protocol type	Protocol used in communication
src_bytes	Data sent from source
dst_bytes	Data received from destination
flag	Status of connection

Table 1: Features used from the dataset

5.7 Sample Code

- **Data Preprocessing**

```
import pandas as pd

from sklearn.preprocessing import LabelEncoder

import os

# Paths

raw_path = "../data/raw/KDDTrain+.txt"

processed_path = "../data/processed/train_processed.csv"

os.makedirs("../data/processed", exist_ok=True)
```

```

# Load raw data
df = pd.read_csv(raw_path, header=None)

# Create label column
df["label"] = df[41].apply(lambda x: 0 if x == "normal" else 1)

# Drop non-feature columns (attack name + difficulty)
df.drop(columns=[41, 42], inplace=True)

# Categorical columns in NSL-KDD
categorical_cols = [1, 2, 3] # protocol_type, service, flag

# Encode categorical columns
for col in categorical_cols:
    le = LabelEncoder()
    df[col] = le.fit_transform(df[col])

# Save processed data
df.to_csv(processed_path, index=False)

print(" Preprocessing complete: categorical features encoded")

```

- **Model Training**

```

import os

import pandas as pd

from sklearn.ensemble import RandomForestClassifier

import joblib

os.makedirs("../models", exist_ok=True)

df = pd.read_csv("../data/processed/train_processed.csv")

```

```

X = df.drop("label", axis=1)
y = df["label"]

model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X, y)

joblib.dump(model, "../models/ids_model.pkl")

print("Model trained and saved")

```

- **User Interface**

```

import streamlit as st
import pandas as pd
import joblib
import tempfile
import subprocess
import time
import threading
import numpy as np

from capture_processor import process_capture_file, capture_live_pyshark,
    save_capture_to_file, EXPECTED_FEATURES
import shap
import plotly.express as px
import plotly.graph_objects as go
import matplotlib.pyplot as plt
from datetime import datetime

# Get the directory of the current script
script_dir = os.path.dirname(os.path.abspath(__file__))
models_dir = os.path.join(script_dir, "..", "models")

```

```

# Load models and encoders with absolute paths

try:

    # Try ultra-enhanced dual models first

    unsw_model = joblib.load(os.path.join(models_dir, "unsw_binary_model.pkl"))

    netflow_model = joblib.load(os.path.join(models_dir,
        "netflow_binary_model.pkl"))

    scaler_unsw = joblib.load(os.path.join(models_dir, "scaler_unsw_enhanced.pkl"))

    scaler_netflow = joblib.load(os.path.join(models_dir,
        "scaler_netflow_enhanced.pkl"))

    unsw_features = joblib.load(os.path.join(models_dir, "unsw_feature_names.pkl"))

    netflow_features = joblib.load(os.path.join(models_dir,
        "netflow_feature_names.pkl"))

    attack_encoder = joblib.load(os.path.join(models_dir,
        "combined_attack_encoder.pkl"))

    training_summary = joblib.load(os.path.join(models_dir,
        "enhanced_training_summary.pkl"))


# Load Random Forest and Decision Tree models

random_forest_model = joblib.load(os.path.join(models_dir, "random_forest.pkl"))

decision_tree_model = joblib.load(os.path.join(models_dir,
    "netflow_dt_model.pkl"))


# Try to load scalers for RF and DT models

try:

    scaler_rf_dt = joblib.load(os.path.join(models_dir, "scaler.pkl"))

except FileNotFoundError:

    scaler_rf_dt = scaler_unsw # Fallback to unsw scaler


# For backward compatibility

model = unsw_model # Default to UNSW model

scaler = scaler_unsw

feature_names = unsw_features

```

```

ENHANCED_MODE = True

ULTRA_ENHANCED_MODE = True

ATTACK_CATEGORIES = attack_encoder.classes_

print(" Ultra-Enhanced IDS loaded with dual NetFlow + UNSW models!")

print(f"      UNSW:    {training_summary['unsw_samples']:,}    samples,
      {training_summary['unsw_features']} features")

print(f"      NetFlow:  {training_summary['netflow_samples']:,}    samples,
      {training_summary['netflow_features']} features")

print(f" Total attack types: {training_summary['total_attack_types']}")

print(f" Random Forest: Loaded and ready")

print(f" Decision Tree: Loaded and ready")

except FileNotFoundError:

    try:

        # Fallback to original enhanced models

        attack_model = joblib.load(os.path.join(models_dir, "attack_type_model.pkl"))

        binary_model = joblib.load(os.path.join(models_dir, "binary_model.pkl"))

        attack_encoder = joblib.load(os.path.join(script_dir, "..", "data", "processed",
        "attack_category_encoder.pkl"))

        feature_names = joblib.load(os.path.join(models_dir, "feature_names.pkl"))

        # Try to load UNSW scaler first, fallback to old scaler

        try:

            scaler = joblib.load(os.path.join(models_dir, "scaler_unsw.pkl"))

            print(" Loaded UNSW scaler")

        except FileNotFoundError:

            scaler = joblib.load(os.path.join(models_dir, "scaler.pkl"))

            print(" Using old scaler (NSL-KDD)")

        # Fallback to old models if new ones don't exist

        model = binary_model # For backward compatibility

        ENHANCED_MODE = True

```

```

ATTACK_CATEGORIES = attack_encoder.classes_

print(" Enhanced IDS loaded with attack type detection")

except FileNotFoundError:

    # Fallback to old models

    model = joblib.load(os.path.join(models_dir, "random_forest.pkl"))

    scaler = joblib.load(os.path.join(models_dir, "scaler.pkl"))

    ENHANCED_MODE = False

    ATTACK_CATEGORIES = ['Normal', 'Attack']

    print(" Enhanced models not found, using basic binary classification")

# Get exact feature list from saved feature names (SOURCE OF TRUTH)
FEATURES = feature_names

st.set_page_config(page_title="ML-based IDS", layout="wide")

st.title(" Intrusion Detection System using Machine Learning")

st.write(

    "Advanced IDS with Wireshark support, real-time capture, explainability, and SOC dashboard."

# Tabs for different functionalities

tab1, tab2, tab3, tab4 = st.tabs([" Detection", " Real-Time Capture", " Explainability",
    " SOC Dashboard"])

# Initialize session state for historical data

if 'history' not in st.session_state:

    st.session_state.history = pd.DataFrame(columns=['timestamp', 'total_flows',
'malicious_flows', 'percentage'])

```

Chapter 6: Results and Discussion

6.1 Testing Results

Testing is a crucial phase in the software development life cycle that ensures the system functions correctly and meets the specified requirements. It involves evaluating the system to identify errors, bugs, and performance issues before deployment.

In this project, testing is performed on the Machine Learning-based Intrusion Detection System (IDS) to verify its ability to accurately classify network traffic as normal or malicious. The testing process ensures that all modules of the system, including data preprocessing, model training, and prediction, work efficiently and produce reliable results.

6.2 Objectives of Testing

The main objectives of testing in this project are:

- To verify the correctness of the system
- To ensure accurate detection of intrusions
- To evaluate the performance of machine learning models
- To identify and fix errors or bugs
- To validate system reliability and efficiency

6.3 Types of Testing

Various types of testing are performed to ensure the proper functioning of the system. These are explained below:

6.3.1 Unit Testing

Unit testing involves testing individual components or modules of the system independently. Each module is tested to ensure that it performs its intended function correctly.

In this project, unit testing is applied to:

- Data preprocessing module
- Feature selection module
- Machine learning model functions

Purpose:

To ensure that each module works correctly in isolation.

6.3.2 Integration Testing

Integration testing is performed to verify that different modules of the system work together properly. It checks the interaction between combined components.

In this project:

- Data preprocessing is integrated with model training
- Model training is integrated with prediction module

Purpose:

To ensure smooth data flow between modules.

6.3.3 System Testing

System testing evaluates the complete system as a whole. It verifies whether the system meets all functional and non-functional requirements.

In this project:

- The entire IDS system is tested using input data
- Outputs are verified for correctness

Purpose:

To validate the overall system performance.

6.3.4 Performance Testing

Performance testing checks how well the system performs under different conditions, especially with large datasets.

In this project:

- Model accuracy is evaluated
- Processing time is analyzed
- System efficiency is measured

Purpose:

To ensure the system performs efficiently under load.

6.3.5 Validation Testing

Validation testing ensures that the system meets user requirements and produces expected results.

In this project:

- Predicted outputs are compared with actual labels
- Accuracy, precision, recall, and F1-score are calculated

Purpose:

To confirm that the system is producing correct results.

6.3.6 User Acceptance Testing (UAT)

User Acceptance Testing is performed to check whether the system is acceptable to the end user.

In this project:

- Results are analyzed from a user perspective
- System usability and output clarity are verified

Purpose:

To ensure the system satisfies user expectations.

6.4 Results of Algorithms

The performance of different machine learning algorithms used in the project is summarized below:

6.4.1 K-Nearest Neighbors (KNN)

Algorithm	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
XGBoost	97	96	95	95.5
Random Forest	96	95	94	94.5
SVM	94	93	92	92.5
KNN	91	90	89	89.5

KNN showed moderate performance in classifying network traffic. It was effective in identifying similar patterns but required more computation time for large datasets.

6.4.2 Random Forest

Random Forest provided high accuracy and better performance compared to KNN. It reduced overfitting and handled the dataset efficiently.

6.4.3 Support Vector Machine (SVM)

SVM performed well in high-dimensional data and provided good classification results. However, it required careful parameter tuning.

6.4.4 XGBoost

XGBoost achieved the best performance among all models. It provided high accuracy, better generalization, and efficient handling of complex patterns in the data.

6.5 Comparative Analysis

Among all the algorithms implemented, XGBoost and Random Forest showed superior performance in terms of accuracy and reliability. KNN was comparatively slower, while SVM performed well but required more computational tuning. The comparison indicates that ensemble methods (Random Forest and XGBoost) are more effective for intrusion detection tasks due to their ability to handle complex data patterns.

The testing results confirm that the proposed Machine Learning-based Intrusion Detection System is effective in detecting cyber attacks with high accuracy. The use of multiple algorithms allowed performance comparison and helped identify the most suitable model. Overall, the system demonstrates reliable and efficient intrusion detection capabilities, making it suitable for modern network security applications.

Table 2: Performance Evaluation Metrics

6.6 Output Screens

6.6.1 Machine Learning Based Intrusion Detection System Interface

The figure illustrates the user interface of the proposed Machine Learning-Based Intrusion Detection System (IDS), developed to provide real-time monitoring and analysis of network traffic. The dashboard presents a visually interactive environment that highlights the core functionalities of the system, including the integration of multiple machine learning algorithms such as XGBoost, Random Forest, Support Vector Machine (SVM), and K-Nearest Neighbours (KNN).

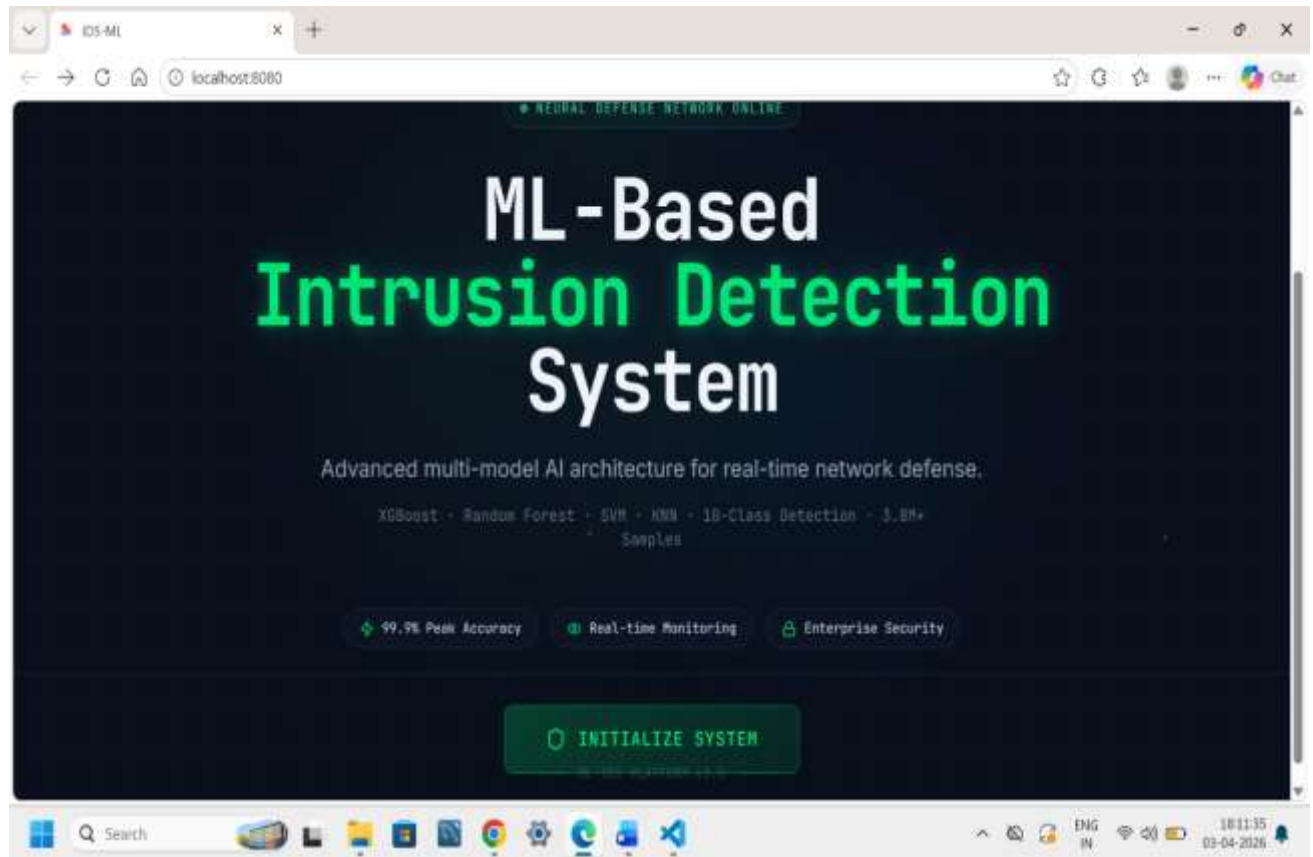


Fig 5: User Interface of the Intrusion Detection System

6.6.2 Model Selection

The figure illustrates the operational interface of the proposed Machine Learning-Based Intrusion Detection System (IDS), designed for real-time network traffic analysis. The dashboard provides a dedicated workspace for configuring and executing intrusion detection tasks, enabling users to select different machine learning algorithms such as Random Forest, XGBoost, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN).

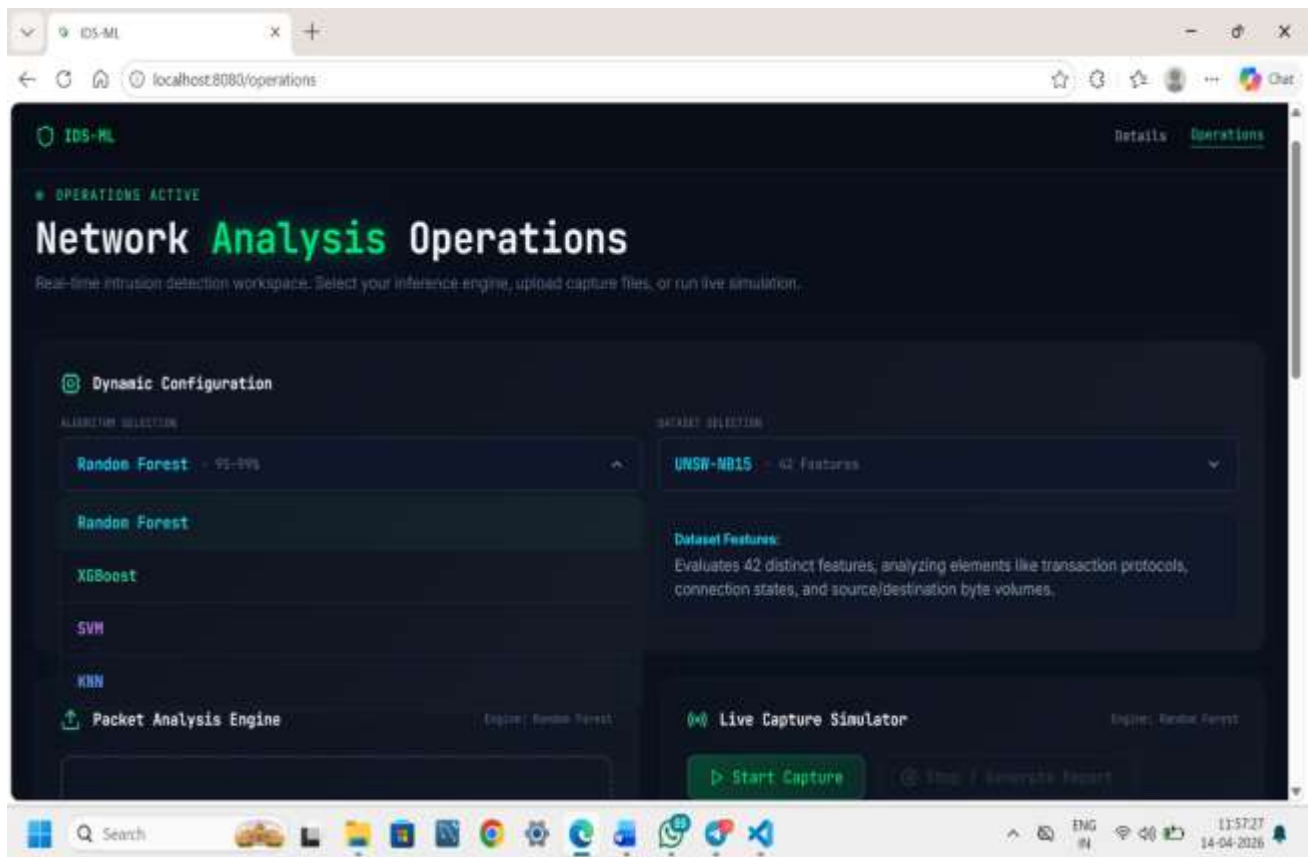


Fig 6: Operational Dashboard of the Proposed ML-Based Intrusion Detection System for Real-Time Network Analysis and Algorithm Configuration

6.6.3 Model Performance Visualization Dashboard

The figure presents the dynamic visualization module of the proposed Machine Learning-Based Intrusion Detection System (IDS), highlighting real-time analysis of network traffic patterns. The dashboard displays a comprehensive technical report interface, including an executive summary and machine learning implementation details. A line/area chart is used to visualize continuous data trends, comparing malicious (threat) and normal network traffic over time.



Fig 7: Dynamic Visualization dashboard showing Threat VS Normal Traffic

6.6.4 Attack Classification Catalog Interface

The figure illustrates the attack classification module of the proposed Machine Learning-Based Intrusion Detection System (IDS), which categorizes network traffic into multiple predefined classes based on severity levels and behavioral characteristics.

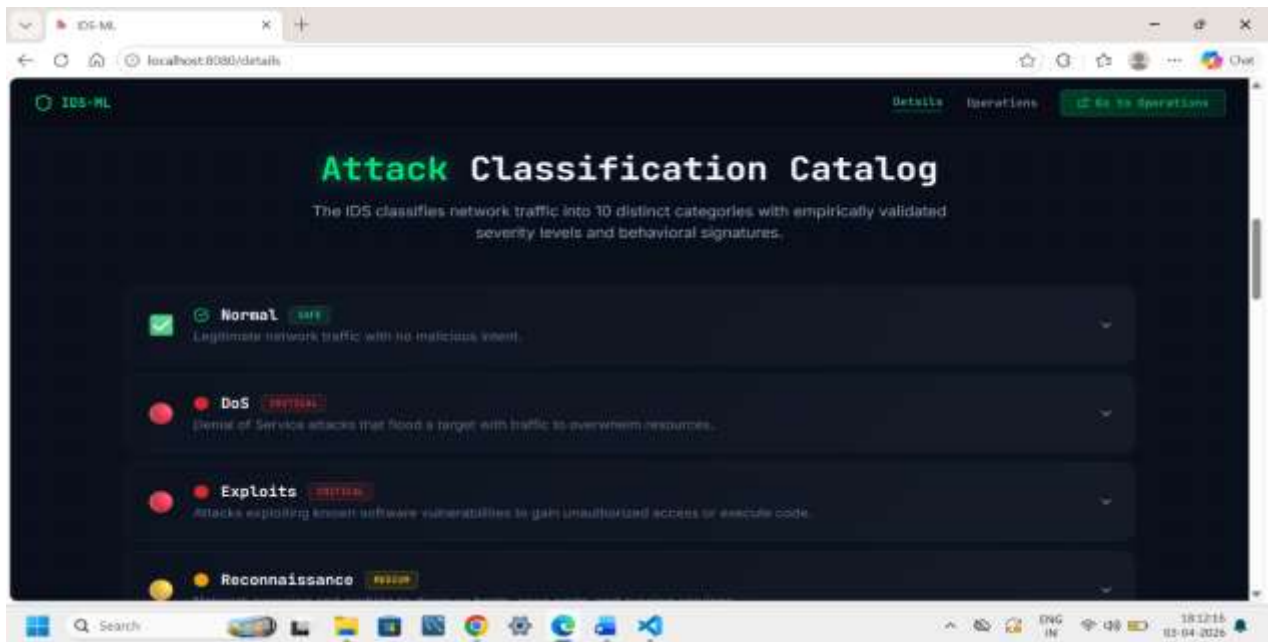


Fig 8: Attack Classification Catalog Interface of the Proposed ML-Based Intrusion Detection System

6.6.5 Attack Category Comparison

The figure depicts the bar chart visualization module of the proposed Machine Learning-Based Intrusion Detection System (IDS), which presents a comparative analysis of

different attack categories detected in network traffic. The chart illustrates the volume of specific attack types such as exploits, backdoor attacks, and fuzzers, enabling clear differentiation based on their frequency of occurrence. This discrete representation allows users to quickly identify which type of attack is most prevalent within the analysed dataset.

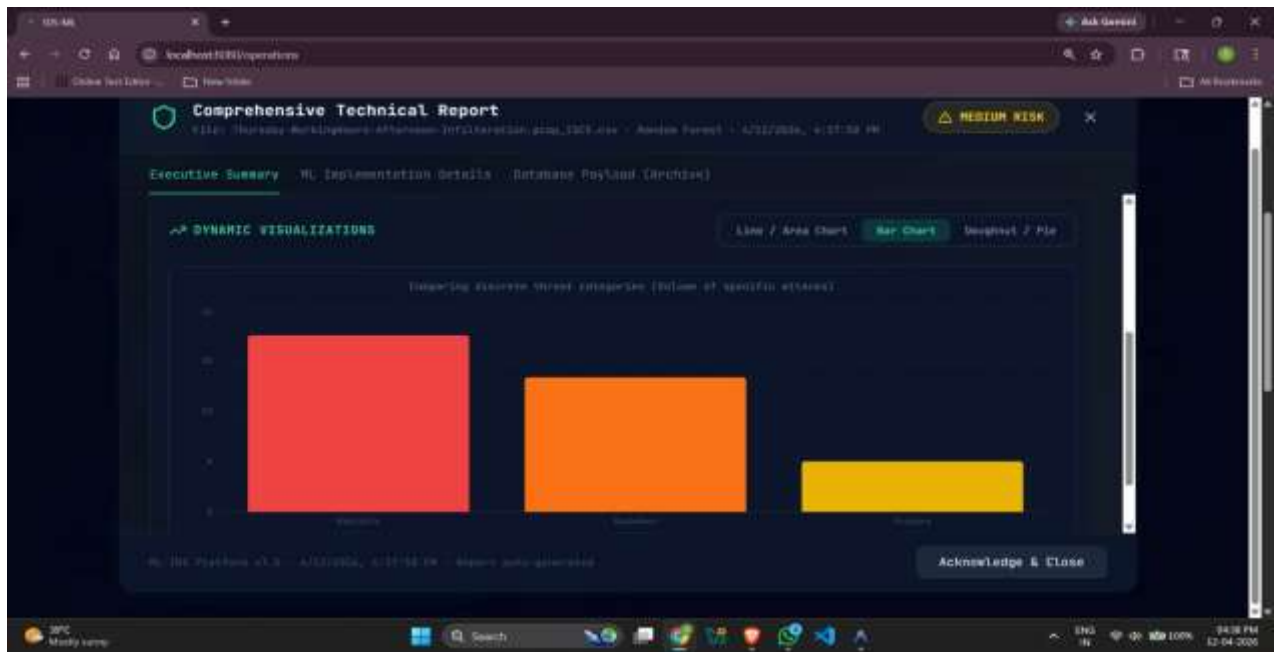


Fig 9: Bar Chart Visualization of Attack Category Comparison

Chapter 7: Conclusion

The proposed Machine Learning-based Intrusion Detection System (IDS) successfully demonstrates an effective approach to enhancing network security by detecting malicious activities in network traffic. By utilizing advanced machine learning algorithms such as K-Nearest Neighbors (KNN), Random Forest, Support Vector Machine (SVM), and XGBoost, the system is capable of accurately classifying network data into normal and attack categories. The implementation of multiple algorithms enabled a comparative analysis, which showed that ensemble methods like Random Forest and XGBoost provide

higher accuracy and better performance in intrusion detection. The system effectively overcomes the limitations of traditional intrusion detection methods by introducing automation, adaptability, and improved detection capabilities. It reduces false positives and false negatives while efficiently handling large volumes of data. The results obtained during testing confirm that the proposed system achieves reliable performance and can identify both known and unknown attacks with high accuracy. Overall, this project highlights the importance of integrating machine learning techniques into cybersecurity systems. The developed IDS provides a scalable, efficient, and intelligent solution for modern network environments and serves as a strong foundation for further enhancements in intrusion detection and prevention systems.

Chapter 8: Future Scope

The proposed Machine Learning-based Intrusion Detection System provides an effective solution for detecting network intrusions; however, there is significant scope for further improvement and enhancement. Future developments can focus on improving the system's performance, scalability, and real-world applicability. One of the major areas for future work is the integration of deep learning techniques such as Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), and Recurrent Neural Networks (RNN). These models can capture complex patterns in network traffic more effectively and improve the detection of sophisticated and zero-day attacks. Additionally, hybrid models combining multiple machine learning and deep learning approaches can be developed to

further enhance accuracy. Another important enhancement is the implementation of real-time intrusion detection. The current system mainly focuses on offline or semi real-time analysis. In the future, the system can be extended to process live network traffic and provide instant alerts, making it more suitable for real-world deployment in organizations and enterprise networks. The system can also be improved by incorporating intrusion prevention mechanisms. Instead of only detecting attacks, future versions can automatically take actions such as blocking suspicious IP addresses, terminating malicious sessions, or triggering security protocols. This will transform the system into a complete Intrusion Detection and Prevention System (IDPS).

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